

## Chapter 26 General Plumbing Requirements: 2023 FBC - Residential, 8<sup>th</sup> edition | UpCodes

### Chapter 26 General Plumbing Requirements

#### Section P2601 General

##### P2601.1 Scope

The provisions of this chapter shall govern the installation of plumbing not specifically covered in other chapters applicable to plumbing systems. The installation of plumbing, *appliances*, *equipment* and systems not addressed by this code shall comply with the applicable provisions of the *Florida Building Code, Plumbing*.

##### P2601.2 Connections to Drainage System

Plumbing fixtures, drains, appurtenances and *appliances* used to receive or discharge liquid wastes or sewage shall be directly connected to the sanitary drainage system of the building or premises, in accordance with the requirements of this code. This section shall not be construed to prevent indirect waste connections where required by the code.

**Exception:** Bathtubs, showers, lavatories, clothes washers and laundry trays shall not be required to discharge to the sanitary drainage system where such fixtures discharge to systems complying with Sections P2910 and P2911.

##### P2601.3 Flood Hazard Areas

In flood hazard areas as established by Table R301.2(1), plumbing fixtures, drains, and *appliances* shall be located or installed in accordance with Section R322.1.6.

#### Section P2602 Individual Water Supply and Sewage Disposal

##### P2602.1 General

The water-distribution and drainage system of any building or premises where plumbing fixtures are installed shall be connected to a public water supply or sewer system, respectively, if available. Where either a public water supply or sewer system, or both, are not available, or connection to them is not feasible, an individual water supply or individual (private) sewage-disposal system, or both, shall be provided. Individual water supplies shall be constructed and installed in accordance with the applicable state and local laws. Where such laws do not address all of the requirements set forth in NGWA-01, individual water supplies shall comply with NGWA-01 for those requirements not addressed by state and local laws.

##### P2602.2 Flood-Resistant Installation

In flood hazard areas as established by Table R301.2(1):

1. Water supply systems shall be designed and constructed to prevent infiltration of floodwaters.
2. Pipes for sewage disposal systems shall be designed and constructed to prevent infiltration of floodwaters into the systems and discharges from the systems into floodwaters.

#### Section P2603 Structural and Piping Protection

##### P2603.1 General

In the process of installing or repairing any part of a plumbing and drainage installation, the finished floors, walls, ceilings, tile work or any other part of the building or premises that must be changed or replaced shall be left in a safe structural condition in accordance with the requirements of the building portion of this code.

##### P2603.2 Drilling and Notching

Wood-framed structural members shall not be drilled, notched or altered in any manner except as provided in Sections R502.2, R602.3, R802.7 and R802.7.1. Holes in load-bearing members of cold-formed steel light-frame construction shall be made only in accordance with Sections R505.1, R603.1 and R804.1. In accordance with the provisions in Sections R505.1, R603.1 and R804.1, cutting

and notching of flanges and lips of load-bearing members of cold-formed steel light-frame construction shall be prohibited. Structural insulated panels (SIPs) shall be drilled and notched or altered in accordance with the provisions of Section R613.7.

#### **P2603.2.1 Protection Against Physical Damage**

In concealed locations, where piping, other than cast-iron or galvanized steel, is installed through holes or notches in studs, joists, rafters or similar members less than 1<sup>1</sup>/<sub>4</sub> inches (31.8 mm) from the nearest edge of the member, the pipe shall be protected by steel shield plates. Such shield plates shall have a thickness of not less than 0.0575 inch (1.463 mm) (No. 16 Gage). Such plates shall cover the area of the pipe where the member is notched or bored, and shall extend not less than 2 inches (51 mm) above sole plates and below top plates.

#### **P2603.3 Protection Against Corrosion**

Metallic piping, except for cast iron, ductile iron and galvanized steel, shall not be placed in direct contact with steel framing members, concrete or masonry. Metallic piping shall not be placed in direct contact with corrosive soil. Where sheathing is used to prevent direct contact, the sheathing material thickness shall be 0.010 inch (0.254 mm) and shall be made of plastic. Where sheathing protects piping that penetrates concrete or masonry walls or floors, the sheathing shall be installed in a manner that allows movement of the piping within the sheathing.

**Exception:** Sleeving is not required for installation of CPVC into concrete or similar material.

#### **P2603.3.1 Penetration**

Protective sleeves around piping penetrating concrete slab-on-grade floors shall not be of cellulose-containing materials. If soil treatment is used for subterranean termite protection, the sleeve shall have a maximum wall thickness of 0.010 inch, and be sealed within the slab using a noncorrosive clamping device to eliminate the annular space between the pipe and the sleeve. No termiticides shall be applied inside the sleeve.

#### **P2603.4 Pipes Through Foundation Walls**

A pipe that passes through a foundation wall shall be provided with a relieving arch, or a pipe sleeve shall be built into the foundation wall. The sleeve shall be two pipe sizes greater than the pipe passing through the wall.

#### **P2603.5 Freezing**

In localities having a winter design temperature of 32°F (0°C) or lower as shown in Table R301.2(1) of this code, a water, soil or waste pipe shall not be installed outside of a building, in exterior walls, in *attics* or crawl spaces, or in any other place subjected to freezing temperature unless adequate provision is made to protect it from freezing by insulation or heat or both. Water service pipe shall be installed not less than 12 inches (305 mm) deep and not less than 6 inches (152 mm) below the frost line.

#### **P2603.5.1 Sewer Depth Reserved.**

### **Section P2604 Trenching and Backfilling**

#### **P2604.1 Trenching and Bedding**

Where trenches are excavated such that the bottom of the trench forms the bed for the pipe, solid and continuous load-bearing support shall be provided between joints. Where over-excavated, the trench shall be backfilled to the proper grade with compacted earth, sand, fine gravel or similar granular material. Piping shall not be supported on rocks or blocks at any point. Rocky or unstable soil shall be over-excavated by two or more pipe diameters and brought to the proper grade with suitable compacted granular material.

#### **P2604.2 Water Service and Building Sewer in Same Trench**

Where the water service piping and *building sewer* piping is installed in same trench, the installation shall be in accordance with Section P2906.4.1.

#### **P2604.3 Backfilling**

Backfill shall be free from discarded construction material and debris. Backfill shall be free from rocks, broken concrete and frozen chunks until the pipe is covered by not less than 12 inches (305 mm) of tamped earth. Backfill shall be placed evenly on both sides of

the pipe and tamped to retain proper alignment. Loose earth shall be carefully placed in the trench in 6-inch (152 mm) layers and tamped in place.

#### P2604.4 Protection of Footings

Trenching installed parallel to footings and walls shall not extend into the bearing plane of a footing or wall. The upper boundary of the bearing plane is a line that extends downward, at an angle of 45 degrees from horizontal, from the outside bottom edge of the footing or wall.

### Section P2605 Support

#### P2605.1 General

Piping shall be supported in accordance with the following:

1. Piping shall be supported to ensure alignment and prevent sagging, and allow movement associated with the expansion and contraction of the piping system.
2. Piping in the ground shall be laid on a firm bed for its entire length, except where support is otherwise provided.
3. Hangers and anchors shall be of sufficient strength to maintain their proportional share of the weight of pipe and contents and of sufficient width to prevent distortion to the pipe. Hangers and strapping shall be of *approved* material that will not promote galvanic action.
4. Where horizontal pipes 4 inches (102 mm) and larger convey drainage or waste, and where a pipe fitting changes the flow direction greater than 45 degrees (0.79 rad), rigid bracing or other rigid support arrangements shall be installed to resist movement of the upstream pipe in a direction opposite the pipe flow. A change of flow direction into a vertical pipe shall not require the upstream pipe to be braced.
5. Piping shall be supported at distances not to exceed those indicated in Table P2605.1.

**TABLE P2605.1**

#### PIPING SUPPORT

| PIPING MATERIAL                                                                              | MAXIMUM<br>HORIZONTAL<br>SPACING (feet) | MAXIMUM<br>VERTICAL<br>SPACING (feet) |
|----------------------------------------------------------------------------------------------|-----------------------------------------|---------------------------------------|
| ABS pipe                                                                                     | 4                                       | 10 <sup>b</sup>                       |
| Aluminum tubing                                                                              | 10                                      | 15                                    |
| Cast-iron pipe                                                                               | 5 <sup>a</sup>                          | 15                                    |
| Copper or copper alloy pipe                                                                  | 12                                      | 10                                    |
| Copper or copper alloy tubing (1 <sup>1</sup> / <sub>4</sub> inches in diameter and smaller) | 6                                       | 10                                    |
| Copper or copper alloy tubing (1 <sup>1</sup> / <sub>2</sub> inches in diameter and larger)  | 10                                      | 10                                    |
| Cross-linked polyethylene (PEX) pipe, 1 inch and smaller                                     | 2.67 (32 inches)                        | 10 <sup>b</sup>                       |
| Cross-linked polyethylene (PEX) pipe, 1 <sup>1</sup> / <sub>4</sub> inch and larger          | 4                                       | 10 <sup>b</sup>                       |
| Cross-linked polyethylene/aluminum/cross-linked polyethylene (PEX-AL-PEX) pipe               | 2.67 (32 inches)                        | 4 <sup>b</sup>                        |
| CPVC pipe or tubing (1 inch in diameter and smaller)                                         | 3                                       | 10 <sup>b</sup>                       |
| CPVC pipe or tubing (1 <sup>1</sup> / <sub>4</sub> inches in diameter and larger)            | 4                                       | 10 <sup>b</sup>                       |

|                                                                     |                  |                 |
|---------------------------------------------------------------------|------------------|-----------------|
| Lead pipe                                                           | Continuous       | 4               |
| PB pipe or tubing                                                   | 2.67 (32 inches) | 4               |
| Polyethylene of raised temperature (PE-RT) pipe, 1 inch and smaller | 2.67 (32 inches) | 10 <sup>b</sup> |
| Polyethylene of raised temperature (PE-RT) pipe, 1¼ inch and larger | 4                | 10 <sup>b</sup> |
| Polypropylene (PP) pipe or tubing (1 inch and smaller)              | 2.67 (32 inches) | 10 <sup>b</sup> |
| Polypropylene (PP) pipe or tubing (1¼ inches and larger)            | 4                | 10 <sup>b</sup> |
| PVC pipe                                                            | 4                | 10 <sup>b</sup> |
| Stainless steel drainage systems                                    | 10               | 10 <sup>b</sup> |
| Steel pipe                                                          | 12               | 15              |

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm.

- a. The maximum horizontal spacing of cast-iron pipe hangers shall be increased to 10 feet where 10-foot lengths of pipe are installed.
- b. For sizes 2 inches and smaller, a guide shall be installed midway between required vertical supports. Such guides shall prevent pipe movement in a direction perpendicular to the axis of the pipe.

## Section P2606 Penetrations

### P2606.1 Sealing of Annular Spaces

The annular space between the outside of a pipe and the inside of a pipe sleeve or between the outside of a pipe and an opening in a building envelope wall, floor, or ceiling assembly penetrated by a pipe shall be sealed with caulking material or foam sealant or closed with a gasketing system. The caulking material, foam sealant or gasketing system shall be designed for the conditions at the penetration location and shall be compatible with the pipe, sleeve and building materials in contact with the sealing materials.

Annular spaces created by pipes penetrating fire-resistance-rated assemblies or membranes of such assemblies shall be sealed or closed in accordance with the building portion of this code.

## Section P2607 Waterproofing of Openings

### P2607.1 Pipes Penetrating Roofs

Where a pipe penetrates a roof, a flashing of lead, copper, galvanized steel or an *approved* elastomeric material shall be installed in manner that prevents water entry into the building. Counterflashing into the opening of pipe serving as a vent terminal shall not reduce the required internal cross-sectional area of the vent pipe to less than the internal cross-sectional area of one pipe size smaller.

### P2607.2 Pipes Penetrating Exterior Walls

Where a pipe penetrates an exterior wall, a waterproof seal shall be made on the exterior of the wall by one of the following methods:

1. A waterproof sealant applied at the joint between the wall and the pipe.
2. A flashing of an *approved* elastomeric material.

## Section P2608 Workmanship

#### **P2608.1 General**

Valves, pipes and fittings shall be installed in correct relationship to the direction of the flow. Burred ends shall be reamed to the full bore of the pipe.

### **Section P2609 Materials Evaluation and Listing**

#### **P2609.1 Identification**

Each length of pipe and each pipe fitting, trap, fixture, material and device utilized in a plumbing system shall bear the identification of the manufacturer and any markings required by the applicable referenced standards. Nipples created from the cutting and threading of *approved* pipe shall not be required to be identified.

**Exception:** Where the manufacturer identification cannot be marked on pipe fittings and pipe nipples because of the small size of such fittings, the identification shall be printed on the item packaging or on documentation provided with the item.

#### **P2609.2 Installation of Materials**

Materials used shall be installed in strict accordance with the standards under which the materials are accepted and *approved*. In the absence of such installation procedures, the manufacturer's instructions shall be followed. Where the requirements of referenced standards or manufacturer's instructions do not conform to the minimum provisions of this code, the provisions of this code shall apply.

#### **P2609.3 Plastic Pipe, Fittings and Components**

Plastic pipe, fittings and components shall be third-party certified as conforming to NSF 14.

#### **P2609.4 Third-Party Certification**

Plumbing products and materials required by the code to be in compliance with a referenced standard shall be *listed* by a third-party certification agency as complying with the referenced standards. Products and materials shall be identified in accordance with Section P2609.1.

#### **P2609.5 Water Supply Systems**

Water service pipes, water distribution pipes and the necessary connecting pipes, fittings, control valves, faucets and appurtenances used to dispense water intended for human ingestion shall be evaluated and listed as conforming to the requirements of NSF 61.